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Design of a statistical model for estimating manufacturing times on the final assembly line of drawers in a wooden furniture plant in the city of Cuenca

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Summary

Design a statistical model using multiple linear regression to estimate manufacturing times, through the use of Industrial Engineering tools.

Objectives

General Objective

Design a statistical model that allows estimating manufacturing times on the final drawer assembly line, based on current or additional variables, in order to reduce the estimation error.

Specific Objective

Define the variables of the final drawer assembly line assembly process that affect the manufacturing time of the company under study.

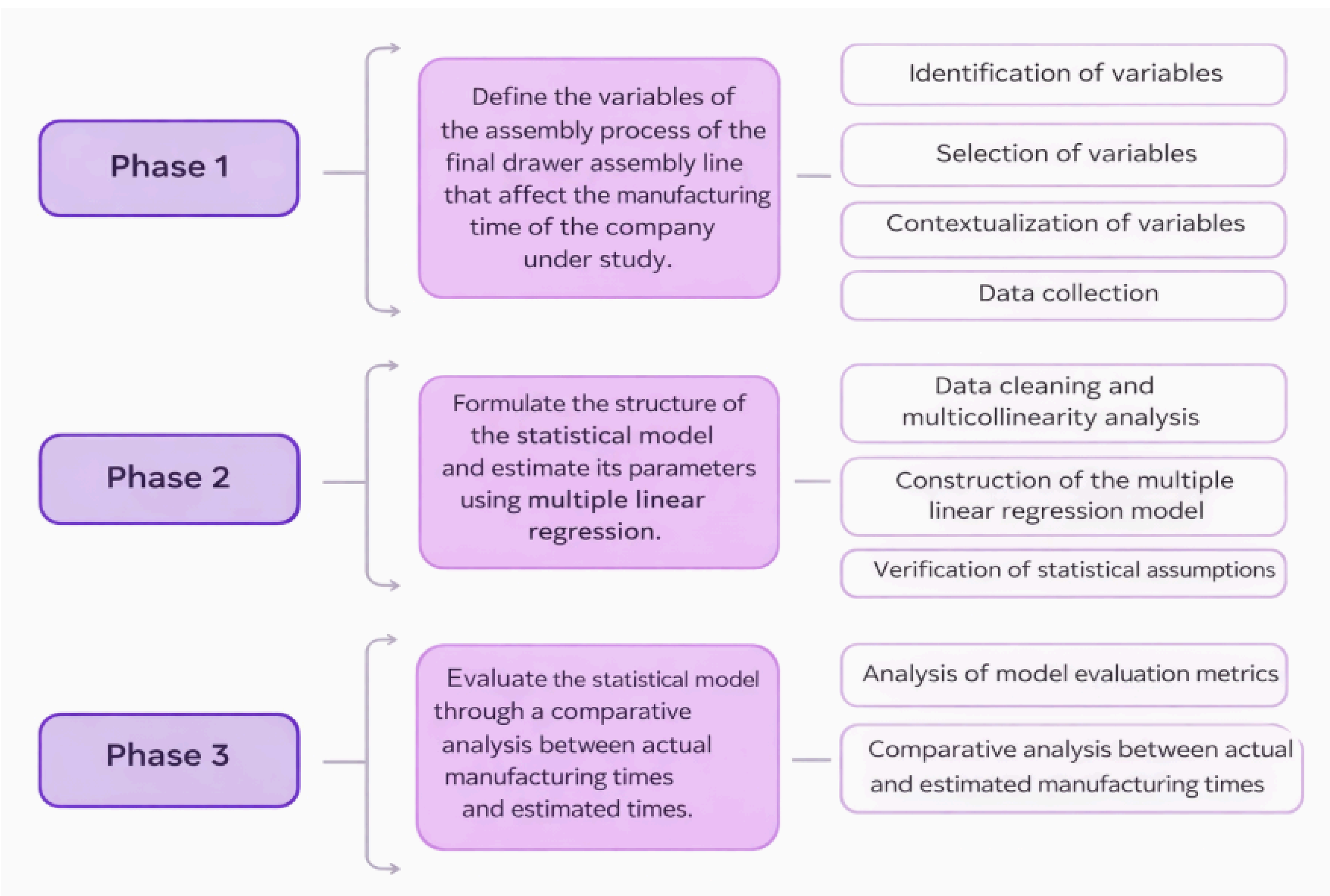
Specific Objective

Formular la estructura del modelo estadístico y estimar sus parámetros mediante regresión lineal múltiple.

Specific Objective

Evaluate the statistical model through a comparative analysis between actual manufacturing times and estimated times.

Methodology



Curriculum Integration

Analytical Statistics.
Design of Experiments (DOE).
Process Engineering and Ergonomics.
Production Organization.
Operations Research.
Production Control Systems.

Applied Engineering

Quality Management Systems: ISO 9001:2015.
Work Studies – Work Measurement: ISO 6385:2016.
Industrial Engineering – Methods and Time Studies: ANSI Z94.0.
Statistical Process Management: ISO 7870.
Good Manufacturing Practices: RTE INEN – BPM.
Occupational Health and Safety: ISO 45001:2018.
Environmental Legislation: TULSMA (Ecuador).

Keywords

Wooden furniture industry

Manufacturing times

Multiple linear regression

Production time estimation

Results

Multiple linear regression models were developed for different furniture families, achieving high levels of fit (R^2 between 0.89 and 0.99) and low estimation errors (MAPE < 6%).

The volume of wood (dm^3) was identified as the main predictor of manufacturing time, especially in product families with standardized products. In families with greater construction complexity, hardware and metal components had a significant impact on assembly time.

The models showed high predictive accuracy, without systematic over- or underestimation biases compared to actual times. In families with low structural variability, simpler models proved sufficient and more parsimonious.

For families with only one product, statistical modeling was not viable; in these cases, the use of actual average times is more appropriate.

Conclusions

Conclusion 1

Multiple linear regression is an effective and reliable tool for estimating manufacturing times in the wooden furniture industry.

Conclusion 2

The selection of variables should be tailored to each product family, avoiding generic approaches. Product size and assembly complexity are the factors that most influence the duration of the production process.

Conclusion 3

Parsimonious models offer better interpretability without sacrificing accuracy.

Conclusion 4

The proposed approach reduces uncertainty in production planning and operational decision-making on the factory floor. The study confirms that interpretable statistical models can be highly competitive with more complex methods when based on real production data.

Contacts

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References

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